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The effects of bidialectalism on cognitive functioning among young and midlife adults in Austria

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Abstract: One of the most widely debated topics in bilingualism research pertains to the issue of the “bilingual advantage”, that is the idea that regular use of multiple languages can enhance cognitive abilities. A related notion is that control of two typologically very close languages such as a vernacular and standard language variety (i.e., bidialectalism) may foster cognitive control abilities due to the ubiquitous everyday usage of both varieties. We set out to investigate to what extent varying usage of and proficiency in standard German and the Austro-Bavarian vernacular correlate with three measures of cognitive functioning, namely working memory, divided attention, and alertness, among 39 young and midlife adults (aged 21–64). We found that, controlling for socio-demographic features, Austro-Bavarian-dominant bidialectals demonstrated superior performance on the working memory task, but not on the alertness or divided attention tasks. These findings shed new light on the effects of bidialectalism on cognitive functioning in adulthood.

Keywords: bidialectalism; bilingualism; cognitive functioning; midlife adults

1 Introduction

The hotly debated “bilingual advantage” – that is, that the use of two (or more) languages in daily life can manifest cognitive benefits (for overviews, see Adesope et al. 2010; Bialystok et al. 2009; Kroll and Bialystok 2013) – has sparked discussions about whether the regular use of a standard language and a regional vernacular variety of the same language (i.e., bidialectalism), might also be linked to advantages in cognitive control (e.g., Alrwaita et al. 2022; Antoniou and Spanoudis 2020; Oswald et al. 2018). Bilingualism scholars and psychologists alike have suggested that bidialectals may need to exert different levels of cognitive control in their daily lives due to the ubiquitous usage necessity of both language varieties as compared to bilinguals who may have a more compartmentalized pattern of language use (e.g., Hsu 2021; Poarch et al. 2019). The plethora of findings about either a bilingual or bidialectal advantage are rather mixed, however, and meta-analyses of the former have reported negligible to small effects after accounting for publication bias (de Bruin et al. 2015; Lehtonen et al. 2018). In light of this controversy, Baum and Titone (2014), among others, question under what circumstances increased cognitive performance emerges and which language environments are conducive to these effects. In order to explore this issue, the focus of the present study is on whether bidialectalism in the Austro-Bavarian dialect region modulates cognitive abilities among 39 young and midlife adults, aged 21–64, who have German as their first language (L1).

2 Background

In their unity/diversity framework, Miyake and Friedman (2012) describe the so-called executive function network which is responsible for individuals’ nonverbal cognitive processes and consists of the subcomponents

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shifting (the ability to switch tasks or mental sets), updating (monitoring and revising working memory), and inhibiting (suppressing impulsive responses). The premise for bilingualism relating to the enhancement of central executive control functions is that, because two or more languages are constantly active in a bilingual's mind (Bialystok et al. 2012), they must constantly inhibit one when speaking the other, depending on the context (Carlson and Meltzoff 2008). This allegedly allows bilinguals to better regulate and control their attention in other (nonlinguistic) cognitive tasks (e.g., Adesope et al. 2010; Bialystok et al. 2016). Bilingual advantages have been reported for numerous cognitive control abilities, such as inhibitory control (Bialystok et al. 2008; Salvatierra and Rosselli 2011), shifting abilities (Prior and MacWhinney 2010; Wiseheart et al. 2016), working memory capacity (Blom et al. 2014; Luo et al. 2013), alertness (e.g., Dash et al. 2019), attentional control (e.g., Bialystok et al. 2008; Zied et al. 2004), and others.

Some researchers cautiously support the idea of a bilingual advantage pertaining to executive functioning (van den Noort et al. 2019) and others challenge it (Paap and Greenberg 2013). Among several of the reasons for contradictory findings about the bilingual advantage, Grundy (2020), for instance, has highlighted the issue of ceiling performance on cognitive tasks in young adults, resulting from the fact that these individuals are in their “cognitive prime years” (Gold et al. 2013: 393). Bak (2016) and van den Berg et al. (2025) stipulate that cognitive effects of bilingualism are more consistently found in older adults. Absence of a bilingual effect in young adults does not imply that there are no consequences of their language experience; rather, their cognitive control functions are operating within a resource-rich system not yet affected by age-related decline. Kroll and Bialystok (2013: 8) poignantly state that “being bilingual as a young adult appears to have a cumulative effect later in life that produces the observed bilingual advantages in the elderly”. In part for this reason, de Bot (2017: 25) has suggested that “rather than one all-purpose bilingual advantage, there may be different advantages – but also disadvantages – for specific groups (early/late bilinguals, migrants, high/low education) and specific cognitive functions (working memory, attention, task switching)”.

While we seem to be making headway in identifying the contexts in which a bilingual advantage manifests, “the cognitive implications of bidialectalism are underexplored” (Ross and Melinger 2017: 14). As pointed out previously, bidialectals' language usage patterns may implicate different cognitive control mechanisms compared to bilinguals due to the potentially differing opportunity in daily life to use both languages. Poarch et al. (2019: 614) argue that “testing bidialectals may offer valuable insights into the frequency of using each language, of switching between the two languages, and its effect on EF [executive functions], given their possibly broader spectrum of usage contexts compared to bilinguals”. Recent years have seen a growing interest in the topic. We find that bidialectals across the lifespan, especially in childhood (Antoniou et al. 2016; Blom et al. 2017; Bosma et al. 2017), young adulthood (Antoniou and Spanoudis 2020; Oschwald et al. 2018; Scaltritti et al. 2017), and older adulthood (Houtzager et al. 2017; Hsu 2021) tend to outperform monolinguals on executive functioning and working memory tasks, though null effects have also been reported (among children: Ross and Melinger 2017; among older adults: Kirk et al. 2014). Particularly relevant to the current investigation are the findings from Poarch et al. (2019), who explored how the differential degree of bidialectalism affects executive functioning among young adults. While they posited that symmetrical use of and proficiency in standard German and the Swabian dialect would correlate with enhanced executive functioning, they instead found that Swabian-dominant bidialectals performed the best on Flanker and Simon tasks (i.e., measures of an individual's ability to inhibit interfering stimuli). They argue that, in the Swabian context, Swabian-dominant bidialectals may represent individuals with more “balanced” bidialectalism, since they are also ubiquitously exposed (e.g., receptively) to standard German in daily life, a nuance which may not have been adequately captured with their bidialectalism questionnaire.

The particular bidialectalism investigated in this paper can be categorized as a dialect–standard continuum, or *diaglossia*, where one can observe a range of varieties between the base dialect (referring to the traditional local vernacular, a regional variety) and the roofing standard language. Unlike in northern and central Germany, the Austro-Bavarian vernacular plays a substantial role in everyday life in Austria, with survey data illustrating that 83 % of Austrians use their local dialect in at least half of their daily (but primarily oral and informal) interactions (Ender and Kaiser 2009). Vernacular forms are also perceptually very distinct from standard variants in terms of phonological, morphological, syntactic, and lexical features (for more extensive descriptions, see Ender and Kaiser 2009; Mauser 2009; Scheutz 2009; Vergeiner 2019; Wiesinger 1983; Wirtz 2025; Zehetner 1985).

Importantly, 87 % of Austrians still report high proficiency in the standard language, emphasizing its ubiquity (if only largely receptively or in written form) in day-to-day activities.

This study addresses several gaps and areas for improvement in bidialectalism research:

- We can consider it an axiom at this point that bilingualism and bidialectalism are not binary but rather radial, continuous, and multidimensional categories (e.g., Luk and Bialystok 2013); individuals are bilingual or bidialectal to different degrees. Grouping bilinguals or bidialectals inherently obscures within-group differences in the relative usage of each language (variety) and in the contexts in which they are used (e.g., Baum and Titone 2014; Oswald et al. 2018; Poarch et al. 2019). In addition to conforming more closely to modern views of bi- and multilingualism (Luk and Bialystok 2013), individual-difference designs such as ours which capitalize on gradient measures of bidialectal use and exposure also substantially reduce the burden of meticulously matching proficiency- and exposure-related variables within each respective group (Poarch and van Hell 2017); that is, bidialectalism can be modeled as a continuous variable rather than as a between-group factor.
- Alrwaita et al. (2022: 848, italics in original) emphasize that “knowing two languages is *not* the key factor in modulating EFs [executive functions]; rather, it is how speakers of a community use each language/variant [sic] in different interactional contexts that strengthens EFs” (see also Green and Abutalebi 2013). This implores us to capture bidialectalism as a composite of different variables “to allow for a more fine-grained assessment of exposure and usage patterns” (Poarch et al. 2019: 625). In this study, we therefore focus specifically on proficiency-based and usage-based bidialectalism.
- Recently, it has been proposed that bilingualism may have a subtle impact on diverse executive functioning measures (Kroll and Bialystok 2013), while Oswald et al. (2018: 17) concluded for young adults that “bidialectism does not practice common executive control as much as it does processes that are more specific to WM [working memory]”. There seems to be a need for studies that explore multiple domains of cognitive control, including executive functioning and working memory, and the differential effects that bidialectalism might have on these.
- Finally, several scholars have sought to determine to what extent children, young adults, and older adults benefit from bidialectalism, but the middle years ranging from approximately 40 to 60 are yet underrepresented (but see, e.g., Ballarini et al. 2023). The age diversity represented in our sample will allow us to propose some initial insights into the extent to which a bidialectal advantage manifests in an age-heterogeneous group of speakers including those in midlife.

Based on the research lacunae we have outlined, we address the following research question:

To what extent can bidialectalism (based on proficiency- and usage-based variables) explain individual differences in cognitive performance (alertness, divided attention, and working memory) among young and midlife adults?

Following in the steps of assumptions made on the basis of the bilingual advantage, that is that use of two or more languages can enhance cognitive abilities (Adesope et al. 2010; Bialystok et al. 2009; Kroll and Bialystok 2013), we suggest that bidialectals reporting symmetrical proficiency and use of standard German and the Austro-Bavarian vernacular will outperform more standard- or Austro-Bavarian-dominant bidialectals. An opposite prediction may be that Austro-Bavarian-dominant bidialectals represent individuals with more “balanced” bidialectalism given the ubiquitous (also largely receptive) exposure to the standard variety in daily life – similar to the conclusions drawn by Poarch et al. (2019) in the Swabian dialect setting – which could suggest a positive linear correlation between cognitive performance and proficiency in and use of the Austro-Bavarian vernacular variety.

3 Methods

3.1 Participants

We recruited 39 participants for the present study, 12 men and 27 women, aged 21–64 ($M = 38.5$, $SD = 10.8$). All participants were from the Middle Bavarian dialect region in Austria, specifically Salzburg ($n = 33$) and Upper

Austria ($n = 6$). Fifteen participants reported no higher education degree. Participants received 20 euros after completion of the full test battery. We did not collect precise data on the age of acquisition for either the vernacular or the standard variety (for a similar suggestion, see also Poarch et al. 2019), as participants were screened to meet two key inclusion criteria: (a) early and continuous residence in Austria, ensuring exposure to the dialect–standard continuum from a young age, and (b) active use of the vernacular variety in everyday communicative contexts. Given our focus on knowledge- and usage-based factors in bidialectalism (see next section), the exact age of onset for each variety was not deemed critical to our research objectives. Instead, consistent exposure and functional use were considered more relevant indicators for our investigation.¹ This approach is also confluent with arguments made by inter alia Pelham and Abrams (2014: 313), such that a bilingual – and by extension a bidialectal – advantage is “the result of proficient, habitual use of 2 languages and not of developmental changes associated with becoming bilingual during childhood”.

3.2 Tasks

3.2.1 Dialect–standard profile

Based on the principles of the Bilingual Language Profile (Birdsong et al. 2012), participants reported on their current language usage practices – specifically on their use of standard German, the Austro-Bavarian vernacular, and any other languages – with friends, with family, when speaking with oneself, during service encounters, at one’s place of education (e.g., university), and at work (participants only reported on the latter two if they indicated they were currently in schooling or engaged in the workforce). Responses within each of the six communicative contexts had to sum up to 100 %. Figure 1 displays radar plots summarizing both the aggregate usage patterns across the six contexts as well as the individual responses, both of which illustrate that a majority of participants employ the vernacular variety at comparatively high rates, followed by standard German, and, finally, additional languages. To simultaneously account for individual differences in language proficiency, they were asked about their speaking skills in standard German and Austrian dialect on a slider scale from 0 to 100 (standard German: $M = 83.3$, $SD = 21.4$, range = 20.0–100.0; Austrian dialect: $M = 92.8$, $SD = 9.7$, range = 60.0–100.0). We then computed a dialect–standard profile (DSP) for each participant (Steiner et al. 2023), such that positive values indicate a higher dominance of the Austro-Bavarian vernacular in terms of everyday language usage and proficiency, negative values a higher dominance of standard German, and values around zero indicating balanced usage- and knowledge-based bidialectalism ($M = 39.9$, $SD = 18.8$, range = –10.5–75.0; see Figure 2).

3.2.2 Cognitive test battery

The present test battery was adopted from Kliesch and Pfenninger (2021), and all tasks were computerized, programmed, and administered using PsychoPy (Peirce 2009). We targeted cognitive domains that are assumed to be fostered by the demands of managing multiple language (varietal) systems. Specifically, bilingualism, and by extension bidialectalism, has been associated with enhancements in divided attentional control, since bilinguals need to constantly coordinate competing languages (e.g., Bialystok et al. 2008; Zied et al. 2004); in working memory, as managing two languages requires holding and manipulating relevant linguistic and contextual information while inhibiting interference from the nontarget language (e.g., Adesope et al. 2010; Grundy and Timmer 2017); and in certain subcomponents of attention such as alertness, since bilinguals may develop heightened sensitivity and responsiveness to incoming stimuli due to the need to rapidly detect and adapt to language cues across different interactional contexts (e.g., Dash et al. 2019). This tripartite framework, we argue, is well suited to disentangle the subtle cognitive signatures of bidialectalism in young and midlife adults. For more detailed descriptions of the tasks, see Kliesch and Pfenninger (2021). Note also that reaction times (RTs) were

¹ Originally, we recruited 41 participants, but two participants did not meet the two key inclusion criteria and were therefore excluded from the analyses reported here.

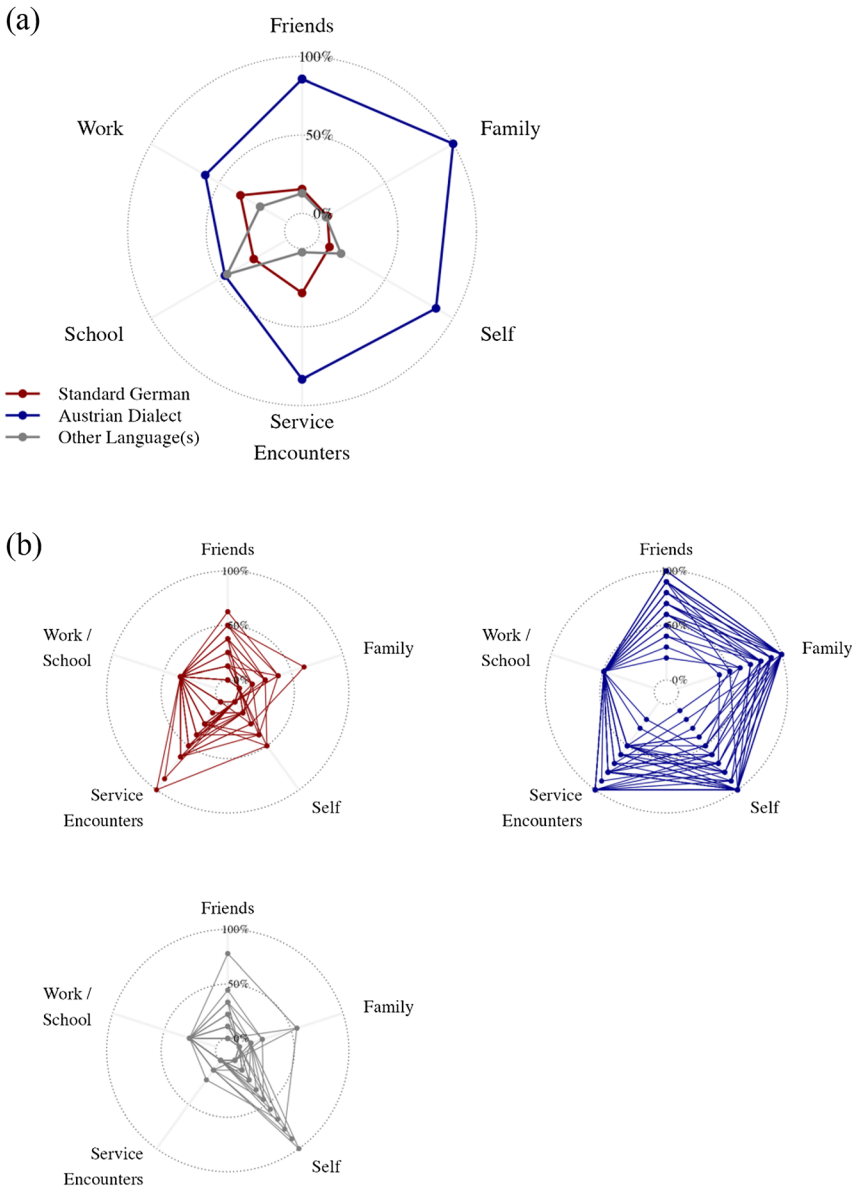


Figure 1: Varietal usage patterns. (a) Aggregate and (b) individual usage patterns across the six contexts. In the individual radar plots in (b), the work and school contexts were combined given that not all participants reported on both contexts. Note that “school” refers to one’s place of education such as university.

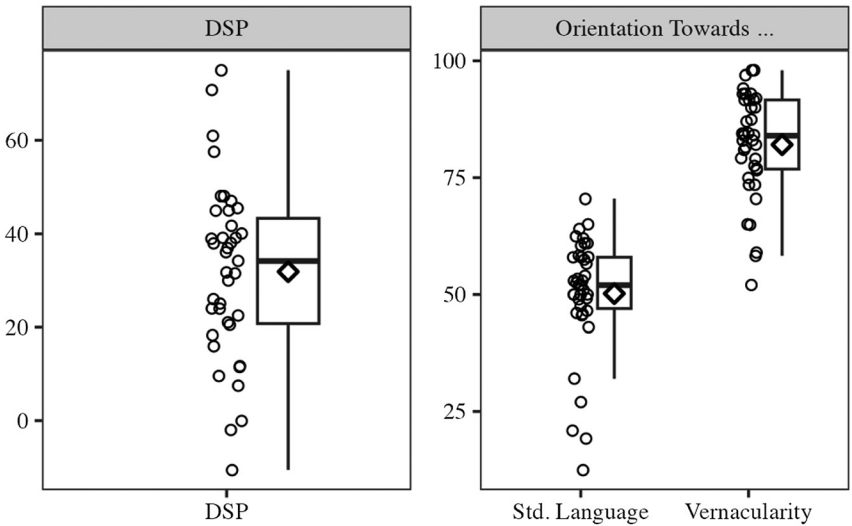


Figure 2: DSP scores and individual orientation towards the standard language and Austro-Bavarian vernacular. The diamond represents the mean.

subtracted from 0 so that variables could be interpreted consistently with positive values indicating better performance.

- Divided attention: The ability to attend to two different stimuli and respond to multiple task demands simultaneously was assessed via a simultaneity test. Participants tracked with the mouse a dot jumping across the screen in intervals of 2 s. Simultaneously, participants were presented with a color word (red, green, or blue) and asked to press a key if the color of the word matched the color the word was printed in (Stroop 1935). Since most participants performed near ceiling level in terms of accuracy, our primary measure is RTs (i.e., the time it took participants to press the key when color and word matched, and only while the mouse was positioned on top of the dot; $M = -1.08$, $SD = 0.24$).
- Working memory: Participants' complex working memory, that is, their ability to store information while being faced with an additional processing task, was measured via an operation span task (Lewandowsky et al. 2010). In this task, participants reported on the correctness of simple math equations (e.g., $8 + 11 = 14$) and were instructed to memorize the consonant shown after each one. After completing a set of between two and six math-consonant pairs, participants were instructed to type the consonants in the order in which they were initially presented. The primary measure of interest in this task was accuracy of recall of consonant sets in the correct order. The measure of accuracy was computed as the proportion of items recalled correctly, also referred to as partial credit scoring (Conway et al. 2005; $M = 0.87$, $SD = 0.12$).
- Alertness: With the term “alertness” we refer to participants' general state of wakefulness that allows them to react swiftly and appropriately to the immediate concrete requirements in their surroundings. This task was adapted from the *Testbatterie zur Aufmerksamkeitsprüfung* or Test of Attentional Performance (TAP; Zimmermann and Fimm 2002). Participants were instructed to press a button as quickly as possible whenever a cross appeared at the center of the computer screen, which appeared for 300 ms ($M = -0.29$, $SD = 0.02$).

RTs were trimmed (a) by removing RTs slower than 2.5 standard deviations from the mean of each participant and (b) removing RTs shorter than 100 ms (i.e., likely accidental button presses). The visual overview of the cognitive data, after z-transformation (i.e., where quantities are expressed in terms of standard deviations from the mean), is presented in Figure 3.

3.3 Data analysis

To address our research question, we employed linear and nonlinear regression modeling in R (R Core Team 2020). Each cognitive domain (alertness, divided attention, and working memory) was modeled separately as a function of the individual DSP scores (i.e., the degree of participants' usage- and proficiency-based bidialectalism). We also account for three control variables: (a) chronological age, so that the resultant model estimates would

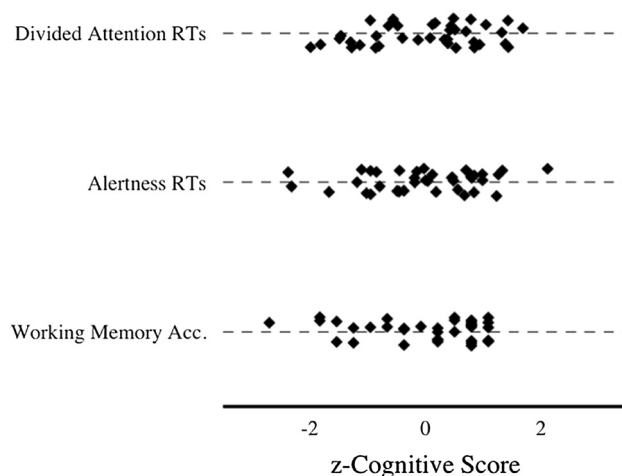


Figure 3: Visual overview of the cognitive variables (z scores).

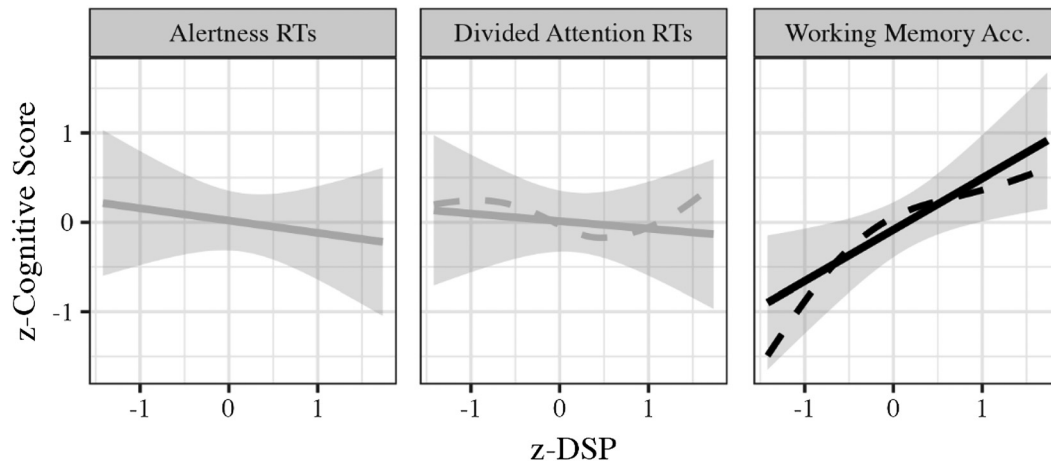


Figure 4: Simple effects of DSP scores on cognitive performance. The solid lines represent the linear regressions, and the dotted lines represent the superimposed generalized additive models.

reflect predictions based on the mean sample age and because processing-based cognitive abilities are known to decline throughout adulthood (e.g., Salthouse 2009, 2012); (b) additional multilingualism, which was assessed via the “other language(s)” item illustrated in Figure 1 and was included because cognitive functioning is thought to be modulated by multilingualism (e.g., Adesope et al. 2010; Bialystok et al. 2009; Kroll and Bialystok 2013); and (c) education, which was entered as a sum-coded binary variable (higher education: 24; no higher education: 15), since higher cognitive abilities are often associated with selection into longer durations of education (Lövdén et al. 2020). A numerical check revealed that the degree of multicollinearity in this particular set of predictors is negligible.

4 Results

Our initial hypothesis presumes the effect of the DSP to be either U-shaped (i.e., more symmetrical proficiency and use correlating with higher cognitive performance) or linear. We thus superimposed a nonlinear regression line via generalized additive models to assess the potentially nonlinear relationship between the DSP scores and cognitive functioning. However, as Figure 4 shows, the linear and nonlinear models appear to be capturing similar general trends. We therefore report on the linear models – since these produce more readily interpretable coefficients – but we report on the generalized additive model in Table 1 in the Appendix. Note that none of the models encountered any convergence issues.

Figure 4 presents the raw correlations (i.e., the uncorrected models before inclusion of our control variables), between DSP scores and our three measures of cognitive performance. We found no relationship between the degree of bidialectalism and alertness ($\beta = -0.14 \pm 0.24$, $t = -0.58$, $p = 0.56$) or divided attention ($\beta = -0.08 \pm 0.24$, $t = -0.35$, $p = 0.73$) but we did detect a significant relationship for working memory. Contrary to our initial hypothesis, it was the Austro-Bavarian-dominant bidialectals who achieved the highest accuracy scores on the working memory task ($\beta = 0.57 \pm 0.22$, $t = 2.65$, $p = 0.01$). Controlling for sociodemographic differences (age, educational attainment, and additional multilingualism; see Tables 2–4 in the Appendix), we draw the same conclusions; in fact, the relationship between the DSP and working memory strengthened slightly ($\beta = 0.62 \pm 0.25$, $t = 2.47$, $p = 0.02$).

Notably, one participant performed 2.7 standard deviations below the mean on the working memory task (i.e., a potential outlier).² Excluding their data indeed weakens the effect ($\beta = 0.42 \pm 0.26$, $t = 1.66$, $p = 0.11$), although

² We do not believe that the low score on the working memory task was an issue of test failure for this particular participant. The potential “outlier” participant performed above average on the alertness task and approximately at average on the divided attention

the effect size remains relatively high and the directionality remains the same. That said, excluding data based solely on standard deviations is not generally recommended (e.g., Nicklin and Plonsky 2020), which is why we employed binary logistic mixed-effect regression using the unaggregated working memory task data as the outcome variable. This approach was also chosen because logistic regression produces robust coefficient estimates even in the face of unbalanced datasets (e.g., Peduzzi et al. 1996). Results from the logistic mixed-effects regression substantiate our conclusions based on the partial scoring (i.e., accuracy) data, both when including the data from the potential “outlier” participant ($\beta = 0.62 \pm 0.26$, $t = 2.42$, $p = 0.02$) and when excluding them ($\beta = 0.48 \pm 0.28$, $t = 1.72$, $p = 0.08$). Taken together, these findings point towards the general trend that Austro-Bavarian-dominant bidialectals demonstrate superior working memory performance, but the sensitivity analysis does urge us to interpret these findings with due caution.

5 Discussion and conclusion

We tested 39 bidialectals (after excluding two participants who did not fit the initial criteria) who varied in terms of their use of and proficiency in standard German and the Austro-Bavarian vernacular variety on three measures of cognitive performance, namely alertness, divided attention, and working memory. We questioned the extent to which the perpetual cognitive challenge of controlling two typologically close languages ubiquitously used in a bidialectal’s language environment might be related to enhancements in domain-general cognitive performance, similar to what has been proposed for bilinguals who speak two more typologically distinct languages.

The main thrust of our study was, at least in part, cautiously substantiated, in that bidialectalism in the Austro-Bavarian context does not seem to be irrelevant when it comes to differences in cognitive functioning. That said, our finding is perhaps counterintuitive at first view: the Austro-Bavarian-dominant bidialectals, that is, those individuals who used the vernacular variety to a greater degree in daily life, tended to demonstrate higher working memory performance than those who used standard and nonstandard varieties in a more balanced manner. Intuitively, and based on what we know from bilingualism research (e.g., Adesope et al. 2010; Bialystok et al. 2009; Kroll and Bialystok 2013), one might expect more balanced bidialectals (as judged via the DSP scores) to more frequently juggle the two varieties and thus demonstrate enhanced executive functioning skills. Poarch’s et al. (2019) findings regarding Swabian bidialectals’ performance on inhibitory control tasks point in a similar direction as ours do, and they provide a reasonable interpretation, with which we agree. Given the widespread presence of the standard variety in public domains such as education, media, and institutional communication, speakers are ubiquitously exposed to diverse forms of standard language. Consequently, cognitive advantages associated with bidialectalism may stem less from symmetrical use of two varieties per se and more from the cumulative cognitive demands of maintaining and using a socially and functionally marked vernacular variety across contexts of everyday use in addition to the consistent, and likely largely receptive, exposure to and engagement with an additional variety such as standard language.

In addition to providing a (post hoc) partial conceptual replication of Poarch’s et al. (2019) findings, our study adds several novel elements to the bidialectalism literature and to the field of bilingualism more broadly. Our finding on the relationship between working memory and higher usage rates of and proficiency in the vernacular variety conform with Oswald’s et al. (2018: 17) proposal based on data from young-adult bidialectals of Swiss standard German and Swiss dialect that bidialectalism “does not practice common executive control as much as it does processes that are more specific to WM [working memory]”, particularly skills relating to the access and retrieval of currently relevant information. Our findings do, however, run counter to Antoniou and Spanoudis’s (2020: 2055) conclusion based on young-adult Cypriot Greek–Standard Modern Greek bidialectals that “the bidialectal advantage ... could not be attributed to a specific EC [executive control] component”. These

task. Relatedly, participants were required to complete a trial run with an experimenter in the room so that any uncertainties could be clarified immediately. Finally, despite the low z score, the participant still achieved 55 % percent accuracy on the working memory task, indicating that they understood the instructions.

contradictory findings, as we see it, emphasize the necessity to continue teasing apart which cognitive control domains bidialectalism affects, and which contextual variables moderate the relationship (e.g., sample age, bidialectal environment). We also highlight that previous meta-analyses on the relationship between bilingualism and working memory capacity have reported significant small to medium effect sizes of $r = 0.20$ (Grundy and Timmer 2017) and $r \approx 0.23$ (Adesope et al. 2010). This hints at the idea that bidialectalism may benefit working memory abilities in a similar way as bilingualism does. Importantly, a sensitivity analysis indicated that our findings can be affected by inclusion or removal of what could (but need not necessarily) be regarded as an outlier (i.e., 2.7 standard deviations below the mean). While the directionality of the effect remained clearly intact, the significance of the effect varied slightly; this suggests a tentative and plausible relationship between bidialectalism and working memory, but calls for studies with a far larger sample size (e.g., via crowdsourcing methods) than we could amass within the confines of this study.

Grundy (2020: 191) calls for us to invest our energy determining “*when* behavioral differences appear [among bilinguals] ... rather than *if* they do”, and we would argue the same holds for bidialectals. Given the sample size and design of the current study, it was not possible to address issues of time and timing concerning when a bidialectal advantage manifests or how variable it is across the lifespan. Prior research has more consistently reported bilingual advantages in older adults (e.g., Bak 2016; Gold et al. 2013; Kroll and Bialystok 2013), likely due to younger adults being in their cognitive prime and therefore performing near ceiling on many cognitive measures, potentially obscuring group-level differences. Kroll and Bialystok (2013) suggest that cognitive effects of bilingualism are cumulative and may only become discernible as individuals traverse (mid-)adulthood. In cognitive and developmental psychology, midlife has been argued to mark a crossroad characterized by both a wealth of accumulated life experiences (Staudinger and Bluck 2001) and the initial emergence of age-related declines in executive functioning (e.g., Salthouse 2012). These features of midlife perhaps make it an informative period for examining *when*, and under what conditions, bilingual or bidialectal cognitive advantages begin to surface, which we offer as an impulse for future (preferably longitudinal) research.

Appendix

Table 1: Results of the generalized additive model.

Parametric coefficients	Estimate	Std. error	<i>t</i>	Pr(> <i>t</i>)
(Intercept)	0.02	0.16	0.11	0.92
z-Divided attention RT	0.00	0.22	0.01	0.99
z-Working memory Acc.	−0.03	0.22	−0.14	0.89
Smooth terms	edf	Ref.df	<i>F</i>	<i>p</i>
z-DSP × z-alertness RT	1.00	1.00	0.38	0.54
z-DSP × z-divided attention RT	2.25	2.82	0.93	0.54
z-DSP × z-working memory Acc.	1.90	2.40	3.09	0.03*

Table 2: Results from the full alertness linear model.

	Estimate ± SE	<i>t</i>	Pr(> <i>t</i>)
(Intercept)	0.02 ± 0.18	0.13	0.89
z-DSP	−0.23 ± 0.27	−0.84	0.40
z-Age	−0.31 ± 0.17	−1.80	0.08
z-Additional multilingualism	−0.25 ± 0.20	−1.25	0.22
z-Educational attainment	−0.13 ± 0.38	−0.34	0.74

Table 3: Results from the full divided attention linear model.

	Estimate ± SE	t	Pr(> t)
(Intercept)	−0.06 ± 0.17	−0.37	0.71
z-DSP	−0.10 ± 0.26	−0.38	0.71
z-Age	−0.21 ± 0.17	−1.28	0.21
z-Additional multilingualism	−0.48 ± 0.20	−2.40	0.02
z-Educational attainment	0.25 ± 0.37	0.68	0.50

Table 4: Results from the full working memory linear model.

	Estimate ± SE	t	Pr(> t)
(Intercept)	−0.12 ± 0.16	−0.79	0.44
z-DSP	0.62 ± 0.25	2.47	0.02*
z-Age	−0.26 ± 0.16	−1.62	0.11
z-Additional multilingualism	−0.16 ± 0.18	−0.84	0.40
z-Educational attainment	0.20 ± 0.36	0.56	0.58

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